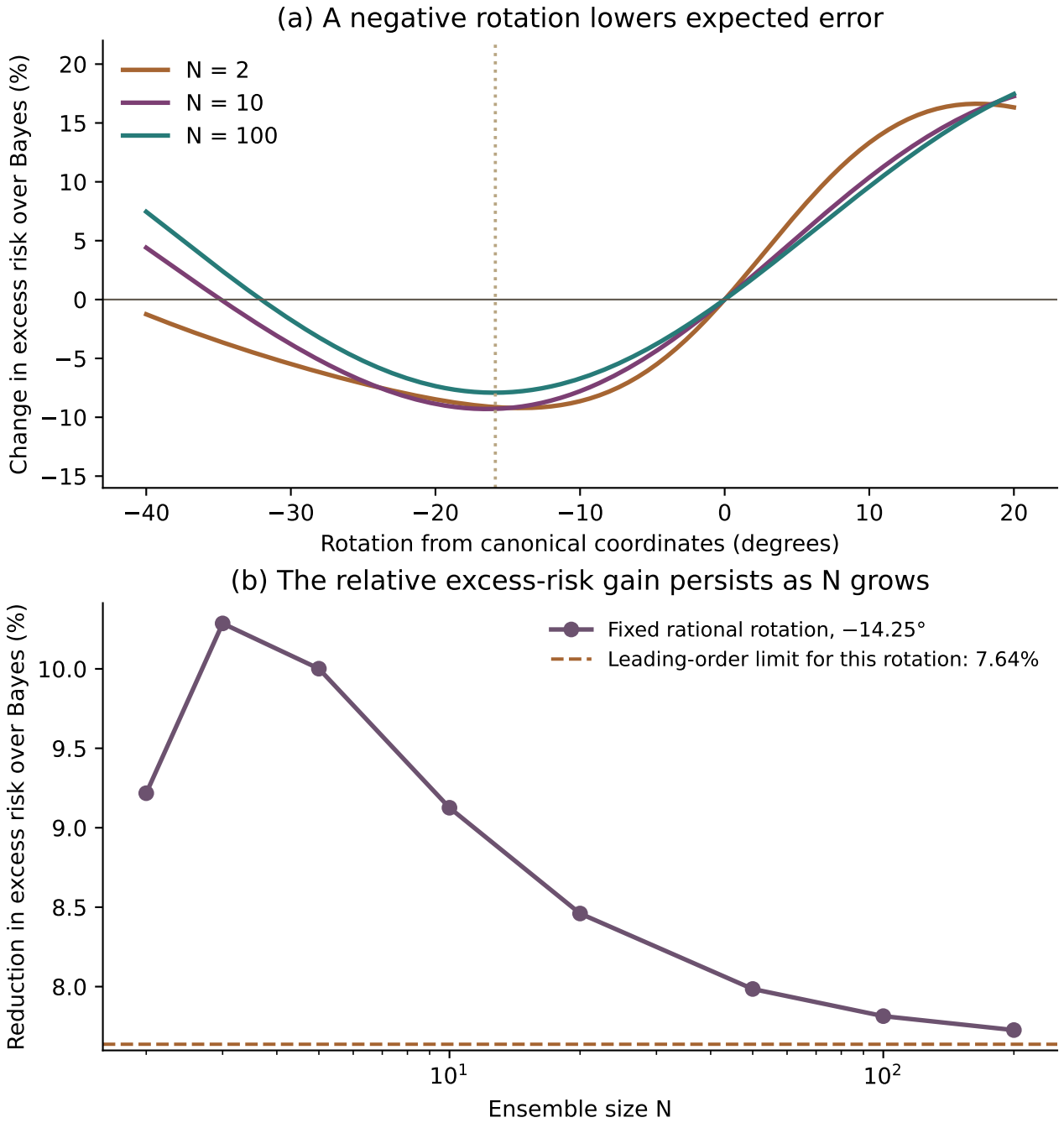
(a) A negative rotation lowers expected error
N = 2
N = 10
N = 100
Change in excess risk over Bayes (%)
20
15
10
5
0
−5
−10
−15
−40
−30
−20
−10
0
10
20
Rotation from canonical coordinates (degrees)
(b) The relative excess-risk gain persists as N grows
Fixed rational rotation, −14.25°
Leading-order limit for this rotation: 7.64%
Reduction in excess risk over Bayes (%)
10.0
9.5
9.0
8.5
8.0
10^1
10^2
Ensemble size N